

Question	Answer	Marks
5(a)	the ratio of stress to strain	B1
5(b)(i)	$A = FL / Ex$	C1
	$A = \text{e.g. } (500 \times 0.81) / (95 \times 10^9 \times 4.0 \times 10^{-3})$	C1
	$A = 1.1 \times 10^{-6} \text{ m}^2$	A1
	OR $A = kL / E$ or $A = \text{gradient} \times L / E$ $k = \text{e.g. } 500 / 4.0 \times 10^{-3}$ $k = 1.25 \times 10^5$	(C1)
	$A = 1.25 \times 10^5 \times 0.81 / 95 \times 10^9$	(C1)
	$A = 1.1 \times 10^{-6} \text{ m}^2$	(A1)

Question	Answer	Marks
5(b)(ii)	$E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} Fx$ or $E = \text{area (under graph)}$	C1
	$(\Delta)E = \frac{1}{2} \times 1.25 \times 10^5 \times ((3.0 \times 10^{-3})^2 - (2.0 \times 10^{-3})^2)$ or $(\Delta)E = (\frac{1}{2} \times 375 \times 3.0 \times 10^{-3}) - (\frac{1}{2} \times 250 \times 2.0 \times 10^{-3})$ or $(\Delta)E = \frac{1}{2} \times (375 + 250) \times 1.0 \times 10^{-3}$	C1
	work done = 0.31 J	A1

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